

LATACUNGA, Ecuador

WMO: 841230

Lat: 0.907S

Lon: 78.616W

Elev: 9205

StdP: 10.42

Time Zone: -5.00 (W05)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	37.2	40.0	33.4	39.5	49.1	35.9	43.7	48.7	25.2	61.7	23.6	61.7	2.7	10	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.4	73.2	55.3	71.5	55.7	69.9	55.4	59.0	66.4	58.0	65.9	57.3	65.4	11.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.3	99.3	59.6	55.7	93.8	58.9	55.1	91.8	58.6	30.5	65.5	29.8	66.3	29.3	64.6	65.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.9	21.1	19.7	DB	31.3	75.8	2.3	1.3	29.7	76.8	28.3	77.5	27.1	78.3	25.4	79.3
			WB	30.5	61.2	2.5	1.8	28.7	62.5	27.2	63.6	25.8	64.6	23.9	65.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	58.0	58.2	57.6	57.4	57.0	55.9	55.0	54.7	55.1	57.3	57.8	57.7
	DBStd	2.68	2.28	2.30	2.11	2.28	2.28	2.00	2.34	2.46	2.74	2.75	2.56	2.39
	HDD50	5	0	0	0	0	0	0	1	1	2	1	0	0
	HDD65	2988	217	192	228	228	247	274	311	318	296	237	215	225
	CDD50	2491	248	228	237	222	218	176	155	148	156	228	235	240
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	12	2	2	0	0	0	0	0	0	0	2	5	1
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	8.6	8.3	7.9	7.1	7.2	7.7	9.9	11.2	11.0	10.0	7.6	6.8	8.0
Precipitation	PrecAvg	28.5	2.4	2.8	3.5	3.7	2.5	1.8	1.0	0.9	1.8	2.6	2.9	2.6
	PrecMax	36.5	6.1	4.9	5.4	6.2	5.8	3.4	1.8	2.0	4.4	5.3	5.2	4.3
	PrecMin	23.1	0.6	0.9	1.8	1.1	0.9	0.7	0.5	0.5	0.7	1.0	1.1	0.8
	PrecStd	4.1	1.3	1.0	0.9	1.2	1.1	0.7	0.4	0.4	0.8	0.9	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.6	73.5	71.9	71.5	71.8	69.4	69.6	70.2	71.6	73.6	74.9	73.4
		MCWB	55.8	56.4	56.8	57.0	55.6	55.3	53.9	53.5	54.6	54.4	54.4	56.3
	2%	DB	71.4	70.7	69.6	68.3	68.7	66.5	66.6	67.7	68.6	71.5	72.0	70.4
		MCWB	55.9	56.2	56.0	56.1	55.6	54.7	53.5	52.8	53.4	55.1	55.0	55.8
	5%	DB	68.9	68.3	67.1	66.4	66.4	64.6	64.7	65.2	66.5	69.4	69.8	68.3
		MCWB	55.4	55.6	55.7	55.9	55.6	54.4	53.1	52.8	53.2	54.8	55.6	55.2
	10%	DB	66.3	66.1	64.7	64.5	64.5	62.8	62.7	63.0	64.6	66.3	66.6	66.1
		MCWB	55.1	55.3	55.2	55.2	55.0	53.7	52.5	52.1	52.8	54.5	54.9	54.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.9	59.5	59.3	59.3	58.5	57.4	56.5	56.1	58.0	58.9	59.4	59.3
		MCDB	67.7	67.6	64.4	64.8	65.7	64.4	65.1	65.1	65.4	66.3	68.3	68.3
	2%	WB	57.5	58.0	58.0	58.0	57.2	56.3	55.0	54.8	55.7	57.1	57.6	57.5
		MCDB	66.5	66.3	64.2	64.6	64.9	63.4	63.3	63.3	64.5	66.6	66.8	66.5
	5%	WB	56.6	56.7	56.9	57.0	56.4	55.3	54.1	53.9	54.5	55.9	56.6	56.4
		MCDB	65.3	64.8	63.9	64.0	64.0	62.4	62.3	62.2	63.3	65.3	65.5	65.2
	10%	WB	55.7	55.9	56.0	56.0	55.6	54.5	53.1	53.0	53.5	54.9	55.7	55.4
		MCDB	63.9	63.6	63.1	62.9	63.1	61.3	60.8	60.9	62.3	63.7	64.1	63.6
Mean Daily Temperature Range	5% DB	MDBR	19.3	18.4	17.7	17.3	17.3	16.2	18.0	19.3	21.5	21.2	21.9	20.0
		MCDBR	23.5	22.3	20.8	19.7	20.2	18.7	21.0	22.8	25.2	25.4	26.8	23.8
		MCWBR	10.0	9.8	9.5	9.2	9.3	8.7	9.8	10.6	12.2	11.3	12.3	10.9
	5% WB	MCDBR	19.9	19.2	18.2	17.7	17.7	16.3	18.4	19.9	21.8	21.6	21.9	20.8
		MCWBR	9.1	8.8	9.3	9.1	8.8	8.4	9.0	9.7	11.0	10.2	10.6	9.8
Clear-Sky Solar Irradiance	taub	0.325	0.349	0.336	0.331	0.313	0.295	0.297	0.332	0.351	0.353	0.351	0.343	
	taud	2.539	2.457	2.513	2.535	2.593	2.660	2.638	2.479	2.416	2.435	2.446	2.480	
	Ebn at Noon	319	312	312	307	305	307	308	303	303	306	308	312	
	Edh at Noon	34	38	36	34	30	28	29	35	38	38	37	36	
All-Sky Solar Radiation	RadAvg	1199	1124	1110	1159	1136	1135	1199	1270	1303	1300	1276	1184	
	RadStd	130	92	71	48	80	74	86	99	100	110	128	76	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page